

Solving Modular Linear Equations

1 The Problem

Goal

We want to solve equations of the form

$$ax \equiv b \pmod{n}$$

where $a > 0$, $n > 0$, and x is the unknown.

This means

$$n \mid (ax - b).$$

So we are looking for all residues x modulo n that make ax have the same remainder as b .

Important

A modular linear equation may have

no solution, one solution, or more than one solution.

The gcd of a and n decides what happens.

2 The Key Number

Let

$$d = \gcd(a, n).$$

The number d measures how much a and n have in common.

Solvability Test

$$ax \equiv b \pmod{n} \text{ has a solution iff } d \mid b.$$

where

$$d = \gcd(a, n).$$

Condition	Meaning
$d \nmid b$	No solution.
$d \mid b$	Solutions exist.
$\gcd(a, n) = 1$	Exactly one solution modulo n .
$\gcd(a, n) = d > 1$ and $d \mid b$	Exactly d solutions modulo n .

3 Why the GCD Test Works

Generated Multiples

In $\mathbb{Z}_n$, the values

$$0, a, 2a, 3a, \dots$$

eventually repeat. The set generated by a is exactly

$$\langle a \rangle = \{0, d, 2d, \dots, (n/d - 1)d\}$$

where

$$d = \gcd(a, n).$$

So $ax \equiv b \pmod{n}$ is solvable exactly when b is one of these multiples of d . That is why

$$d \mid b$$

is the test.

4 How Many Solutions?

Number of Solutions

Let

$$d = \gcd(a, n).$$

Then

$$ax \equiv b \pmod{n} \text{ has either no solution or exactly } d \text{ solutions.}$$

If $d \nmid b$, there are no solutions.

If $d \mid b$, there are d distinct solutions modulo n .

5 Finding One Solution

Use Extended Euclid on a and n :

$$\text{ExtendedEuclid}(a, n) = (d, x', y').$$

This means

$$d = ax' + ny'.$$

So

$$ax' \equiv d \pmod{n}.$$

If $d \mid b$, multiply by b/d :

$$a \left(x' \frac{b}{d} \right) \equiv b \pmod{n}.$$

One Solution

$$x_0 = x' \left(\frac{b}{d} \right) \bmod n.$$

6 Finding All Solutions

Once we have one solution x_0 , the full set of solutions is:

All Solutions

$$x_i = x_0 + i \left(\frac{n}{d} \right) \pmod{n}$$

for

$$i = 0, 1, \dots, d - 1.$$

The spacing between consecutive solutions is

$$\frac{n}{d}.$$

7 Algorithm

Modular Linear Equation Solver

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ModularLinearEquationSolver(a, b, n):  
    (d, x_prime, y_prime) = ExtendedEuclid(a, n)  
  
    if d divides b:  
        x0 = x_prime * (b / d) mod n  
  
        for i = 0 to d - 1:  
            print (x0 + i * (n / d)) mod n  
    else:  
        print "no solutions"
```

Algorithm Meaning

First find $d = \gcd(a, n)$. If d does not divide b , stop. If d divides b , compute one solution x_0 , then add n/d repeatedly to get all d solutions.

8 Worked Example

Problem

Solve

$$14x \equiv 30 \pmod{100}.$$

Step 1: Compute the gcd

$$d = \gcd(14, 100) = 2.$$

Since

$$2 \mid 30,$$

solutions exist. There will be exactly

$$\boxed{2}$$

solutions modulo 100.

Step 2: Run Extended Euclid

First run the Euclidean algorithm:

$$100 = 14(7) + 2,$$

$$14 = 2(7) + 0.$$

So

$$\gcd(14, 100) = 2.$$

Now back-substitute to write 2 as a combination of 14 and 100. From

$$100 = 14(7) + 2,$$

we get

$$2 = 100 - 14(7).$$

Rewrite this in the form

$$2 = 14x' + 100y'.$$

So

$$2 = 14(-7) + 100(1).$$

So

$$x' = -7.$$

This is the coefficient of 14, and it is the value used in the formula for x_0 .

Step 3: Compute one solution

Use

$$x_0 = x' \left(\frac{b}{d} \right) \bmod n.$$

Here,

$$a = 14, \quad b = 30, \quad n = 100, \quad d = 2.$$

Therefore,

$$x_0 = (-7) \left(\frac{30}{2} \right) \bmod 100.$$

$$x_0 = (-7)(15) \bmod 100.$$

$$x_0 = -105 \bmod 100.$$

$$\boxed{x_0 = 95.}$$

Step 4: Find all solutions

The spacing is

$$\frac{n}{d} = \frac{100}{2} = 50.$$

Since $d = 2$, use $i = 0, 1$:

$$x_0 + 0(50) = 95,$$

$$x_0 + 1(50) = 145 \equiv 45 \pmod{100}.$$

Answer

$$\boxed{x \equiv 95 \pmod{100}} \quad \text{or} \quad \boxed{x \equiv 45 \pmod{100}}.$$

Check

For $x = 95$:

$$14(95) = 1330 \equiv 30 \pmod{100}.$$

For $x = 45$:

$$14(45) = 630 \equiv 30 \pmod{100}.$$

So both answers are correct.

9 Special Case: Modular Inverse

When $b = 1$

The equation

$$ax \equiv 1 \pmod{n}$$

asks for the multiplicative inverse of a modulo n .

Inverse Rule

$$\boxed{a^{-1} \pmod{n} \text{ exists iff } \gcd(a, n) = 1.}$$

If it exists, the solution is unique modulo n .

10 Final Review

Core Facts

$$d = \gcd(a, n).$$

$ax \equiv b \pmod{n}$ is solvable iff $d \mid b$.

If solvable, there are exactly d solutions modulo n .

$x_0 = x'(b/d) \pmod{n}$, where $d = ax' + ny'$.

$x_i = x_0 + i(n/d) \pmod{n}$, $i = 0, \dots, d-1$.